Theorem 1 (Chebyshev's Inequality)

For a random variable X with mean μ and variance σ^2, for any $k > 0$:

$$P(| X - \mu | \geq k\sigma) \leq \frac{1}{k^2}$$

Apply Markov's inequality to the non-negative random variable $(X - \mu)^2$.

Remark 1. The bound is loose for small k; it becomes meaningful only once $k > 1$.

Solution 1. Use $k = 2$ to conclude at least 75% of the mass lies within two standard deviations of the mean.

The full argument appears in **?@prf-chebyshev**.